



DATABASE MANAGEMENT

ACADEMIC

Grade 12
Curriculum Guide

DATABASE MANAGEMENT
Prerequisite: General Mathematics (Core Subject)

Course Description:

This course equips learners with fundamental concepts of databases and Structured Query Language (SQL), enabling them to write queries for database management and data manipulation, apply SQL functions, and design simple databases.

Elective: Academic

Time Allotment: 80 hours for one term

CONTENT	CONTENT STANDARDS The learners demonstrate an understanding that...	LEARNING COMPETENCIES The learners...
Introduction to Databases and SQL Commands		
1. Introduction to Databases	1. databases and database management systems store, organize, and manage data efficiently, and that relational and nonrelational models serve different purposes in real-world applications across various industries.	1. define database and database management system. 2. differentiate relational and non-relational databases, their purposes and functions. 3. discuss real-world applications of databases and database management system.
2. Introduction to SQL	2. Structured Query Language (SQL) is an essential tool for creating, managing, and manipulating data in databases, and that SQL concepts such as tables, rows, columns, syntax structures, and data types support the effective organization and processing of	4. explain the basic concepts of SQL and its role in data management and analysis. 5. explain the concepts of tables, rows, and columns in a database. 6. create an Entity-Relationship (ER) Diagram representing entities, attributes, and relationships in a relational database.

CONTENT	CONTENT STANDARDS The learners demonstrate an understanding that...	LEARNING COMPETENCIES The learners...
	information.	7. explain the roles of primary keys and foreign keys in establishing and maintaining relationships between database tables. 8. identify key components of SQL syntax and common SQL data types. 9. create databases and tables using SQL, including inserting and updating data.
3. SQL Commands a. Data Querying Commands (Select, Select Distinct, Where, Order By, And, Or, Not, Like, Wildcards, In, Between, Aliases) b. Data Manipulation Commands (Insert Into, Update, Delete) c. Aggregate Functions (Aggregate Functions, Min and Max, Count, Sum, Avg) d. Grouping and Filtering Data (Group By, Having, Exists, Any All) Joins and Sets Operations (Inner, Left, Right, Full, Self, Union)	3. SQL commands enable data to be queried, modified, aggregated, grouped, filtered, and combined across tables, allowing meaningful information to be retrieved from databases through well-structured executable queries.	10. differentiate SQL command structures based on their purpose and usage in data definition, data manipulation, and data querying. 11. write executable queries against a database using an appropriate SQL command.
SQL Commands and Database Management System		

CONTENT	CONTENT STANDARDS The learners demonstrate an understanding that...	LEARNING COMPETENCIES The learners...
4. SQL Commands Advanced SQL Features (Select Into, Insert Into Select, Case) - Stored Procedures and Comments (Stored Procedures, Comments) - Null Handling (Null Values, Null Functions) - Operators (Arithmetic, Comparison, Compound, Logical)	4. the basic concepts of databases and SQL, including queries, functions, and commands, enable effective database management and manipulation.	12. identify the structure of a SQL command statement 13. explain how SQL clauses are used to filter results. 14. write executable queries against a database using appropriate SQL commands. 15. develop a database management system using SQL. 16. present the developed database management system before an audience.
Performance Standards - The learners utilize basic database concepts, SQL functions, and SQL commands to perform database management tasks. - The learners use basic database concepts, SQL functions, and SQL commands to design simple databases.		